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Control Systems

UNIT - II

29/7/15.

TIME RESPONSE ANALYSIS.

Most of the control systems use time as an independent Variable, so it is important to analyse the response given by the system. for the applied excitation which is function of time. This response is called Time Response of control system.

It is divided into two parts. 1. Transient Response
2. Steady state Response.

1. Transient Response :

It is defined as the part of the time response that goes to zero as time goes to infinity. It shows the response of the system when input changes from one state to another.

2. Steady-state Response :

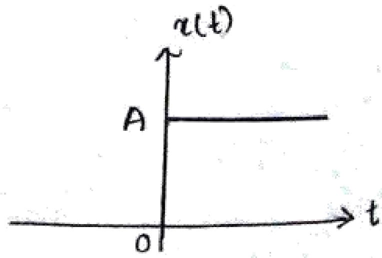
It is a part of total response that remains after transient has died out. It shows the response as t tends to infinity.

Time Response can be obtained by solving differential equations of the system. the time response c(t) can be obtained from transfer function and input to the system.

* Test signals :

Commonly used test input signals are step, impulse, ramp, acceleration, parabolic, sinusoidal etc.

1. Step Input :



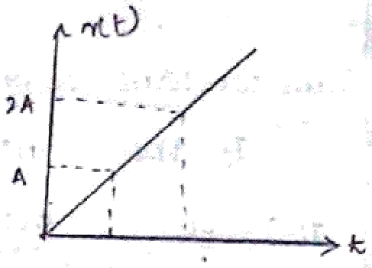
$r(t) = A\,u(t)$

where $u(t) = 1,\ t \geq 0$
$= 0,\ t < 0$

2. Ramp :

$r(t) = At \ , \ t \geq 0$

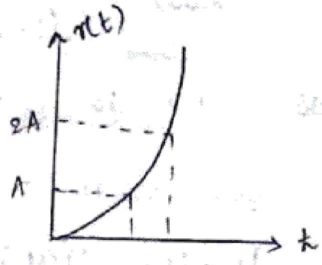
$\quad = 0 \ , \ t < 0$



3. Parabola :

$r(t) = \frac{At^2}{2} \ , \ t \geq 0$

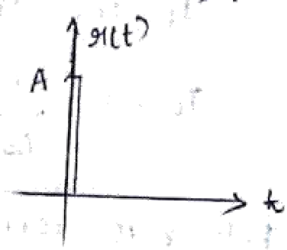
$\quad = 0, \ t < 0$



4. Impulse :

$\delta(t) = 0 \quad for \quad t \neq 0$

$\lim_{t \to 0} \int_{-t_1}^{t_1} \delta(t) \, dt = 1$



Transfer function and Order of a system :

The order of differential equation governing the system gives the order of the system.

It can also be determined from Transfer function

$$TF = K \frac{P(s)}{Q(s)}$$

where K is constant

P(s) is Numerator polynomial

Q(s) is Denominator polynomial.

The order of the system is given by highest power of s in denominator polynomial Q(s), if

$$T(s) = \frac{C(s)}{R(s)} = \frac{b_m s^m + b_{m-1} s^{m-1} + b_{m-2} s^{m-2} + \dots + b_1 s + b_0}{a_n s^n + a_{n-1} s^{n-1} + \dots \dots + a_1 s + a_0}$$

The numerator is a polynomial of degree 'm' and denominator is a polynomial of degree 'n'.

The degree 'n' of denominator polynomial is known as order of the system.

If $n=0$, then the system is zero order system

If $n=1$, then the system is I order system &...

* Time constant Form :

If the transfer function can be represented as

$$T(s) = \frac{C(s)}{R(s)} = \frac{K(1+sT_{z1})(1+sT_{z2})(1+sT_{z3})\cdots}{(1+sT_{p1})(1+sT_{p2})(1+sT_{p3})\cdots}$$, then

this form is known as Time constant form.

Pole-zero form :

If the transfer function is expressed as

$$T(s) = \frac{K(s \pm z_1)(s \pm z_2)\cdots}{(s \pm p_1)(s \pm p_2)\cdots}$$, then it is known as

pole-zero form.

where $z_1, z_2, z_3, \ldots$ are roots of numerator polynomial and are known as zeroes of system

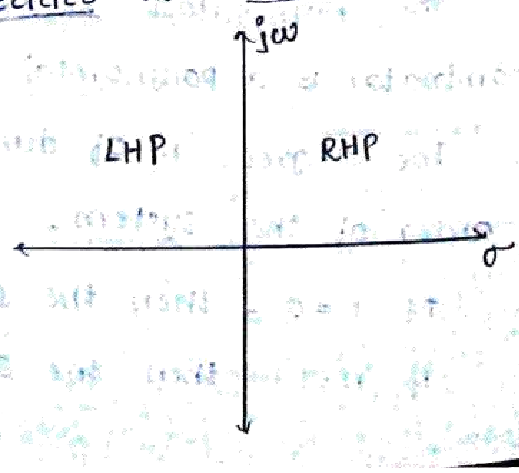
$p_1, p_2, p_3, \ldots$ are roots of denominator polynomial known as poles of the system.

The denominator polynomial of the transfer function is known as CHARACTERISTIC EQUATION.

The roots of characteristic equation are nothing but poles of the system.

- Poles and zeroes may be real (or) complex.
- complex conjugate poles occur in pairs.
- The complex plane i.e; s-plane is used to represent poles and zeroes. The location of poles and zeroes in the s-plane decides the stability of the system

S-plane



The set of all complex coordinates (σ, ω) forms s-plane.

The horizontal axis (σ-axis) is represented by $s = \sigma + j\omega 0$.

The vertical axis ($j\omega$-axis) (imaginary axis) represented by $s = 0 + j\omega$.

- symbol used to indicate a zero in s-plane 'o'.

symbol used to indicate a pole in s-plane 'x'.

* Time Response of First-order system :

The most general form of differential eqn representation of first order system may be expressed as

$$\tau \frac{dy(t)}{dt} + y(t) = K x(t)$$

where $x(t)$ is input variable

$y(t)$ is output variable

K is gain of the system and

τ is time constant of the system.

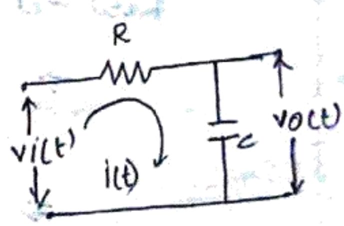
Applying Laplace to the Equation.

$$\tau\, s(Y(s)) + Y(s) = K X(s)$$

$$\boxed{\frac{Y(s)}{X(s)} = \frac{K}{1+s\tau}}$$

This is the transfer function of 1^{st} order.

Example of 1^{st} order system :



Apply KVL to the loop

$$v_i(t) = i(t)R + \frac{1}{C}\int \cancel{vo(t)}\, i(t) \cdot dt$$

Apply Laplace

$$V_i(s) = I(s)R + \frac{1}{Cs} I(s)$$

$$\Rightarrow V_i(s) = I(s)\left[R + \frac{1}{Cs}\right]$$

$$V_o(s) = \frac{1}{C}\int i(t)\, dt$$

$$V_o(s) = \frac{1}{Cs} I(s).$$

$$\frac{V_o(s)}{V_i(s)} = \frac{\frac{1}{Cs}}{R + \frac{1}{Cs}}$$

$$\frac{V_o(s)}{V_i(s)} = \frac{1}{1+RCs}$$

Let $T = RC$

$$\Rightarrow \frac{V_o(s)}{V_i(s)} = \frac{1}{1+sT}$$

consider the transfer function of I order ~~functi~~system

$$\frac{C(s)}{R(s)} = \frac{K}{1+sT} \quad \text{[in time constant form]}.$$

$$\Rightarrow \frac{C(s)}{R(s)} = \frac{K/T}{s+1/T} \quad \text{[this is in pole-zero form]}$$

$$\Rightarrow C(s) = \left[\frac{K}{1+sT}\right]R(s) \quad \text{(or)} \quad C(s) = \left[\frac{K/T}{s+1/T}\right]R(s)$$

$$\Rightarrow c(t) = \mathcal{L}^{-1}\left[\frac{K}{1+sT}R(s)\right]$$

Impulse response of I order system :

$$\frac{C(s)}{R(s)} = \frac{K}{1+sT}$$

$r(t) = \delta(t)$

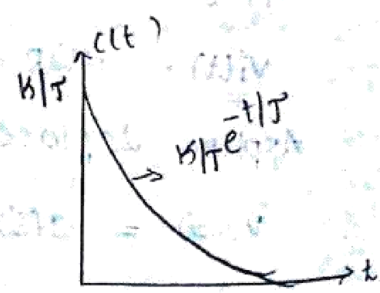
$R(s) = 1$

$$\Rightarrow C(s) = \frac{K}{1+sT} \cdot 1 \quad [\because R(s) = 1].$$

$$\Rightarrow c(t) = \mathcal{L}^{-1}\left[\frac{K}{1+sT}\right]$$

$$\Rightarrow c(t) = \frac{1}{T}\mathcal{L}^{-1}\left(\frac{K}{s+1/T}\right)$$

$$\Rightarrow c(t) = \frac{K}{T}e^{-t/T}$$



Unit step response of I order system :

$$\frac{C(s)}{R(s)} = \frac{K}{1+sT}$$

$r(t) = u(t)$

$R(s) = \frac{1}{s}$

$\Rightarrow C(s) = \frac{K}{1+sT} \cdot R(s)$

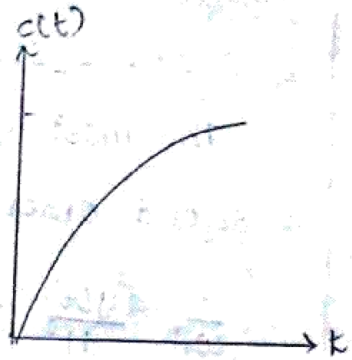
$\Rightarrow C(s) = \frac{K}{1+sT} \cdot \frac{1}{s}$

$\Rightarrow C(s) = \frac{K}{T}\left[\frac{1}{s(s+\frac{1}{T})}\right]$

$\Rightarrow C(s) = \frac{K}{T}\left[\frac{T}{s} - \frac{T}{s+\frac{1}{T}}\right]$

$\Rightarrow c(t) = K\left[\mathcal{L}^{-1}\left(\frac{1}{s} - \frac{1}{s+\frac{1}{T}}\right)\right]$

$\Rightarrow c(t) = K\left[1 - e^{-t/T}\right]$



$\frac{1}{s(s+\frac{1}{T})} = \frac{A}{s} + \frac{B}{s+\frac{1}{T}}$

put $s = 0$

$1 = A(s+\frac{1}{T}) + Bs$

$A = T$

$s = -\frac{1}{T}$

$B = -T$

Response of I order system for ramp input :

Input $r(t) = t$.

$R(s) = \frac{1}{s^2}$.

$\Rightarrow C(s) = \frac{K}{1+sT}\left[\frac{1}{s^2}\right]$

$\Rightarrow C(s) = \frac{K}{T}\left[\frac{1}{(s+\frac{1}{T})s^2}\right]$

$\Rightarrow C(s) = \frac{K}{T}\left[\frac{T^2}{s+\frac{1}{T}} + \frac{T}{s^2} - \frac{T^2}{s}\right]$

$\Rightarrow C(s) = K\left[\frac{T}{s+\frac{1}{T}} + \frac{1}{s^2} - \frac{T}{s}\right]$

$\Rightarrow c(t) = K\left[Te^{-t/T} + t - T\right]$

$\Rightarrow c(t) = K\left[t - T(1-e^{-t/T})\right]$

$\frac{1}{(s+\frac{1}{T})s^2} = \frac{A}{s+\frac{1}{T}} + \frac{B+sC}{s^2}$

$1 = s^2A + (s+\frac{1}{T})(B+sC)$

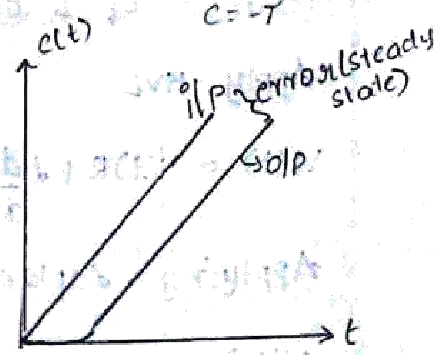
$1 = s^2A + sB + s^2C + \frac{B}{T} + \frac{1}{T}sC$

$A + C = 0$, $B + \frac{1}{T}C = 0$

$A = T^2$

$T + \frac{1}{T}C = 0$ $\frac{B}{T} = 1$

$C = -T^2$ $B = T$.



* II Order systems :

The most general form of differential equations for a second order system may be expressed as

$$\frac{1}{\omega_n^2}\frac{d^2y(t)}{dt^2} + \frac{2\xi}{\omega_n}\frac{dy(t)}{dt} + y(t) = K\,x(t) \longrightarrow (1)$$

where ω_n = natural frequency of the system

K = gain of the system

ξ = damping ratio of the system

$x(t)$ = input variable

$y(t)$ = output variable.

ω_n and K depends on the characteristics of the system under considerations.

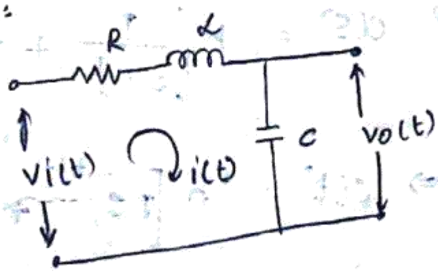
Applying Laplace transform to eqn-(1) & assuming initial conditions to zero.

$$\Rightarrow \frac{1}{\omega_n^2}s^2Y(s) + \frac{2\xi}{\omega_n}sY(s) + Y(s) = KX(s)$$

$$\Rightarrow \frac{Y(s)}{X(s)} = \frac{K}{\frac{1}{\omega_n^2}s^2 + \frac{2\xi}{\omega_n}s + 1}$$

$$\frac{Y(s)}{X(s)} = \frac{K\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$$

Example of II order system :



Apply KVL

$$V_i(t) = i(t)R + L\frac{di}{dt} + \frac{1}{C}\int i(t)\,dt$$

Applying Laplace transform

$$V_i(s) = I(s)R + LsI(s) + \frac{1}{Cs}I(s)$$

$$V_i(s) = I(s)\left[R + Ls + \frac{1}{Cs}\right]$$

$$V_o(t) = \frac{1}{C}\int i(t)\,dt$$

$$\Rightarrow V_o(s) = \frac{1}{Cs} I(s).$$

$$\frac{V_o(s)}{V_i(s)} = \frac{R + \frac{1}{Cs}}{R + Ls + \frac{1}{Cs}}$$

$$\frac{V_o(s)}{V_i(s)} = \frac{\frac{1}{C}}{Rs + Ls^2 + \frac{1}{C}} \quad [\times N \& D \text{ by } s]$$

$$\frac{V_o(s)}{V_i(s)} = \frac{\frac{1}{LC}}{s^2 + s\frac{R}{L} + \frac{1}{LC}} \quad [\div N \& D \text{ with } L]$$

comparing this eqn with standard eqn $\frac{Y(s)}{X(s)}$

$$\Rightarrow \frac{1}{LC} = \omega_n^2 \Rightarrow \omega_n = \frac{1}{\sqrt{LC}}$$

$$\Rightarrow 2\xi\omega_n = \frac{R}{L} \Rightarrow \xi = \frac{R}{2L\omega_n}$$

$$\xi = \frac{R}{2L\left[\frac{1}{\sqrt{LC}}\right]}$$

$$\xi = \frac{R}{2}\sqrt{\frac{C}{L}}.$$

* Impulse Response of Second Order System :

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2+2\xi\omega_n s+\omega_n^2}$$

when $r(t) = \delta(t) \Rightarrow R(s) = 1$

$$\Rightarrow C(s) = \left[\frac{\omega_n^2}{s^2+2\xi\omega_n s+\omega_n^2}\right]$$

case(i) :

when $\xi = 0$, $s_1, s_2 = \pm j\omega_n$.

$$C(s) = \left[\frac{\omega_n^2}{(s^2+\omega_n^2)}\right]$$

$$\frac{\omega_n^2}{s^2+\omega_n^2} = \frac{As+B}{s^2+\omega_n^2} \qquad \Rightarrow c(t) = L^{-1}\left[\frac{\omega_n^2}{s^2+\omega_n^2}\right]$$

$$\omega_n^2 = As+B . \qquad \Rightarrow c(t) = \omega_n \sin\omega_n t \cdot u(t).$$

$$B = \omega_n^2$$

case(ii) : $0 < \xi < 1$

$s_1, s_2 = -\xi\omega_n \pm j\omega_d$, Let $s_a = -s_1$, $s_b = -s_2$.

$$C(s) = \frac{\omega_n^2}{s^2+2\xi\omega_n s+\omega_n^2}[1] .$$

$$\Rightarrow C(s) = \frac{\omega_n^2}{(s+s_a)(s+s_b)}$$

$$\frac{\omega_n^2}{(s+s_a)(s+s_b)} =$$

$$\Rightarrow C(s) = \frac{\omega_n^2}{s^2+2\xi\omega_n s+\omega_n^2} R(s)$$

$$C(s) = \frac{\omega_n^2}{s^2+2\xi\omega_n s+(\xi\omega_n)^2+\omega_n^2-(\xi\omega_n)^2}$$

$$C(s) = \frac{\omega_n^2}{(s+\xi\omega_n)^2+\omega_n^2-(\xi\omega_n)^2}$$

$$C(s) = \frac{\omega_n^2}{(s+\xi\omega_n)^2+\omega_n^2[1-\xi^2]} \qquad \omega_d = \omega_n\sqrt{1-\xi^2}$$

$$c(s) = \frac{\omega_n^2}{(s+\xi\omega_n)^2+\omega_d^2} = \frac{\omega_n \frac{\omega_d}{\sqrt{1-\xi^2}}}{(s+\xi\omega_n)^2+\omega_d^2}$$

Apply Inverse Laplace

$$c(t) = \omega_n \frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}} \cdot \sin\omega_d t$$

iii) $\xi = 1$

$$s_1, s_2 = -\omega_n$$

$$c(s) = \frac{\omega_n^2}{s^2+2\omega_n s+\omega_n^2}.$$

$$c(s) = \frac{\omega_n^2}{(s+\omega_n)^2}$$

$$c(s) = \omega_n^2\left[\frac{1}{(s+\omega_n)^2}\right].$$

$$c(t) = \omega_n^2 \cdot t \cdot e^{-\omega_n t}.$$

iv) $\xi > 1$

$$s_1 s_2 = -\xi\omega_n \pm \omega_n\sqrt{\xi^2-1}$$

$$-s_1 = s_a = \xi\omega_n - \omega_n\sqrt{\xi^2-1}$$

$$-s_2 = s_b = \xi\omega_n + \omega_n\sqrt{\xi^2-1}$$

$$c(s) = \frac{\omega_n^2}{s^2+2\xi\omega_n s+\omega_n^2} \cdot 1 = \frac{A}{s+s_a} + \frac{B}{s+s_b}$$

$$\omega_n^2 = A(s+s_b) + B(s+s_a)$$

at $s = s_a \Rightarrow \omega_n^2 = A(s_b - s_a)$

$$A = \frac{\omega_n^2}{s_b - s_a}$$

at $s = -s_b \Rightarrow \omega_n^2 = B(s_a - s_b)$

$$B = \frac{\omega_n^2}{s_a - s_b}.$$

$$s_a - s_b = \xi\omega_n - \omega_n\sqrt{\xi^2-1} - \xi\omega_n - \omega_n\sqrt{\xi^2-1}$$

$$= -2\omega_n\sqrt{\xi^2-1}$$

$$c(s) = \frac{\omega_n^2}{(s+s_a)(s_b-s_a)} + \frac{\omega_n^2}{(s+s_b)(s_a-s_b)}$$

* Characteristic Equation :

Transfer function of second order system

$$\frac{C(S)}{R(S)} = \frac{\omega_n^2}{S^2 + 2\xi\omega_n S + \omega_n^2}$$

The characteristic equation is $S^2 + 2\xi\omega_n S + \omega_n^2 = 0.$

$$S = \frac{-2\xi\omega_n \pm \sqrt{(\xi\omega_n 2)^2 - 4\omega_n^2}}{2}$$

$$S = -\xi\omega_n \pm \sqrt{(\xi^2-1)\omega_n^2}$$

$$S_1 = -\xi\omega_n + \omega_n\sqrt{\xi^2-1}$$

$$S_2 = -\xi\omega_n - \omega_n\sqrt{\xi^2-1}$$

Effect of ξ on 2nd order system performance

case (i) : $\xi = 0$.

$$\Rightarrow S_1, S_2 = \pm j\omega_n$$

The roots are purely imaginary and the system is undamped.

case (ii) : $0 < \xi < 1$

$$S_1, S_2 = -\xi\omega_n \pm j(\sqrt{1-\xi^2})\omega_n$$

$$= -\xi\omega_n \pm j\omega_d \quad , \quad \text{where } \omega_d = \omega_n\sqrt{1-\xi^2}$$

Roots are complex conjugate and the system is under-damped.

case (iii) : $\xi = 1$

$$S_1, S_2 = -\omega_n.$$

Roots are purely real and equal and the system is critically damped.

case (iv) : $\xi > 1$

$$S_1, S_2 = -\xi\omega_n \pm \omega_n\sqrt{\xi^2-1}$$

Roots are real and unequal and the system is over damped.

* Unit Step Response of Second Order System :

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2+2\xi\omega_n s+\omega_n^2}$$

when $r(t) = u(t) \Rightarrow R(s) = \frac{1}{s}$.

$$C(s) = \left[\frac{\omega_n^2}{s^2+2\xi\omega_n s+\omega_n^2}\right]\left[\frac{1}{s}\right]$$

case (i) :

when $\xi=0$, $s_1, s_2 = \pm j\omega_n$

$$C(s) = \left[\frac{\omega_n^2}{s(s^2+\omega_n^2)}\right].$$

$$\frac{\omega_n^2}{s(s^2+\omega_n^2)} = \frac{A}{s} + \frac{Bs+C}{s^2+\omega_n^2}.$$

$$\omega_n^2 = As^2+A\omega_n^2+Bs^2+Cs.$$

$$C=0 , \quad A+B=0 , \quad A=1.$$

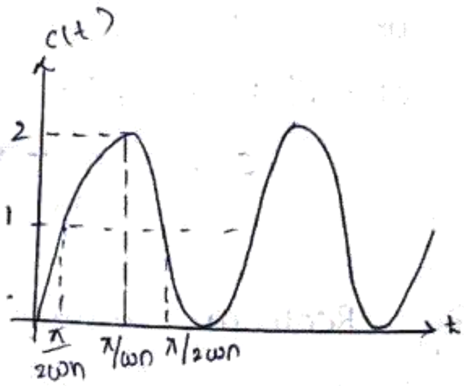
$$1+B=0$$

$$B=-1.$$

$$\Rightarrow C(s) = \frac{1}{s} + \frac{s}{s^2+\omega_n^2}.$$

$$c(t) = \mathcal{L}^{-1}\left(\frac{1}{s} - \frac{s}{s^2+\omega_n^2}\right).$$

$$c(t) = u(t) - \cos\omega_n t \cdot u(t).$$



case

The response of undamped second order system for unit step input is completely oscillatory. Every practical system has some amount of damping. Hence undamped system does not exist in practise.

case (ii) : $0<\xi<1$

$$s_1, s_2 = -\xi\omega_n \pm \sqrt{\;}\, j\omega_d.$$

$$C(s) = \frac{\omega_n^2}{s(s^2+2\xi\omega_n s+\omega_n^2)}$$

$$C(s) = \frac{A}{s} + \frac{Bs+C}{s^2+2\xi\omega_n s+\omega_n^2}.$$

$$\Rightarrow C(s) = \frac{1}{s} - \frac{s+2\xi\omega_n}{s^2+2\xi\omega_n s+\omega_n^2}$$

$\omega_n^2 = As^2 + 2A\xi\omega_n s + A\omega_n^2 + Bs^2 + Cs.$

$A+B=0,\ A=1,$ $B=-1$

$2\xi\omega_n + C = 0$

$C = -2\xi\omega_n$

$$\Rightarrow C(s) = \frac{1}{s} - \frac{s+2\xi\omega_n}{s^2+2\xi\omega_n s+(\xi\omega_n)^2+\omega_n^2-(\xi\omega_n)^2}$$

$$\Rightarrow C(s) = \frac{1}{s} - \frac{s+2\xi\omega_n}{(s+\xi\omega_n)^2+\omega_n^2(1-\xi^2)}$$

$$\Rightarrow C(s) = \frac{1}{s} - \frac{s+2\xi\omega_n}{(s+\xi\omega_n)^2+\omega_d^2} \quad (\because \omega_d = (\sqrt{1-\xi^2})\,\omega_n)$$

$$\Rightarrow C(s) = \frac{1}{s} - \frac{s+\xi\omega_n}{(s+\xi\omega_n)^2+\omega_d^2} - \frac{\xi\omega_n\omega_d}{\omega_d[(s+\xi\omega_n)^2+\omega_d^2]}$$

$$\Rightarrow c(t) = u(t) - \cos\omega_d t \cdot e^{-\xi\omega_n t}\cdot u(t) - \frac{\xi\omega_n}{\omega_d} e^{-\xi\omega_n t}\sin\omega_d t\cdot u(t)$$

$$= 1 - e^{-\xi\omega_n t}\left[\cos\omega_d t + \frac{\xi\omega_n}{\omega_n\sqrt{1-\xi^2}}\sin\omega_d t\right]$$

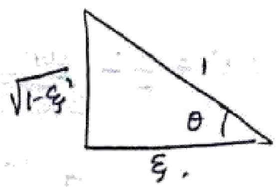
$$= 1 - \frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}}\left[\sqrt{1-\xi^2}\cos\omega_d t + \xi\sin\omega_d t\right]$$

$$= 1 - \frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}}\left[\sin\theta\cos\omega_d t + \cos\theta\sin\omega_d t\right]$$

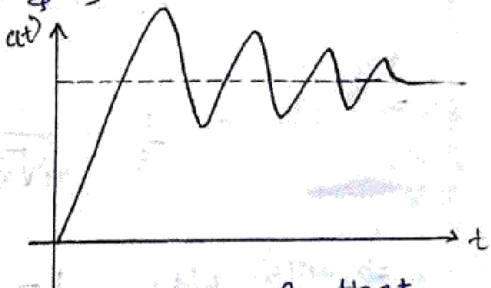
$$c(t) = 1 - \frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}}\sin(\omega_d t+\theta).$$

where $\sin\theta = \sqrt{1-\xi^2}$

$\cos\theta = \xi$



$$\Rightarrow \tan\theta = \frac{\sqrt{1-\xi^2}}{\xi} \Rightarrow \theta = \tan^{-1}\left[\frac{\sqrt{1-\xi^2}}{\xi}\right]$$



The response of under-damped second order system oscillates before settling to final value. The oscillations depends on value of ξ. The frequency of transient oscillations is that damped oscillations natural frequency ω_d ω_n thus varying the damping ratio.

case(iii) : $\xi = 1.$

$s_1, s_2 = -\omega_n.$

$$C(s) = \frac{\omega_n^2}{(s+\omega_n)^2}\frac{1}{s} = \frac{\omega_n^2}{s(s+\omega_n)^2}$$

$$C(s) = \frac{1}{s} - \frac{1}{s+\omega_n} - \frac{\omega_n}{(s+\omega_n)^2}$$

$$\frac{\omega_n^2}{s(s+\omega_n)^2} = \frac{A}{s} + \frac{B}{s+\omega_n} + \frac{C}{(s+\omega_n)^2}$$

$$\omega_n^2 = A(s+\omega_n)^2 + Bs(s+\omega_n) + Cs$$

let $s=0$ $\Rightarrow \omega_n^2 = A(\omega_n^2)$, $\boxed{A=1}$

$A+B=0$, $B=-1$

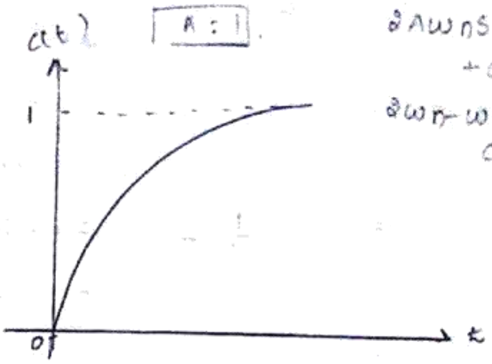
$2A\omega_n s + B\omega_n s + C = 0$

$2\omega_n - \omega_n + C = 0$

$C = -\omega_n$

$$c(t) = 1 - e^{-\omega_n t} - \omega_n t e^{-\omega_n t}$$

$$c(t) = 1 - e^{-\omega_n t}[1+\omega_n t]$$



The response of critically damped second order system has no oscillations.

case(iv) : $\xi > 1$

$$s_1, s_2 = -\xi\omega_n \pm \omega_n\sqrt{\xi^2-1}$$

Let $s_a = -s_1$, $s_b = -s_2$

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2+2\xi\omega_n s+\omega_n^2} = \frac{\omega_n^2}{(s+s_a)(s+s_b)}$$

$$\Rightarrow C(s) = \frac{\omega_n^2}{s(s+s_a)(s+s_b)} = \frac{\omega_n^2}{s_a s_b}\left[\frac{1}{s}\right] - \frac{\omega_n^2}{s_a(s_a+s_b)}\left[\frac{1}{s+s_a}\right] + \frac{\omega_n^2}{s_b(s_b+s_a)(s+s_b)}$$

$\omega_n^2 = A(s+s_a)(s+s_b) + Bs(s+s_b) + Cs(s+s_a)$

$$s_a - s_b = \xi\omega_n + \omega_n\sqrt{\xi^2-1} - \xi\omega_n + \omega_n\sqrt{\xi^2-1}$$

$$= 2\omega_n\sqrt{\xi^2-1}, \quad s_a s_b = (\xi\omega_n - \omega_n\sqrt{\xi^2-1})(\xi\omega_n + \omega_n\sqrt{\xi^2-1})$$

$$= \xi^2\omega_n^2 - \omega_n^2\xi^2 + \omega_n^2$$

$$= \omega_n^2.$$

$s=0$: $\omega_n^2 = A s_a s_b \Rightarrow A = \frac{\omega_n^2}{s_a s_b}$

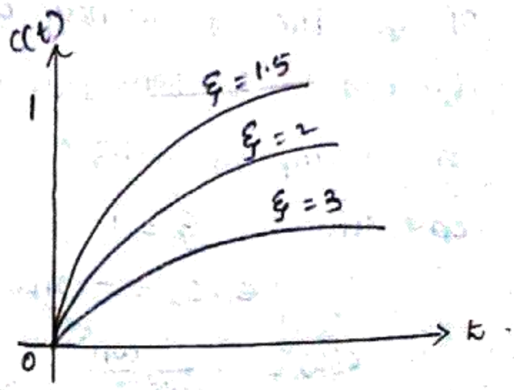
$s = -s_a$: $\omega_n^2 = B(s_b - s_a)s_a$, $B = \frac{-\omega_n^2}{(s_b - s_a)s_a}$

$s = -s_b$: $\omega_n^2 = -s_b(s_a - s_b)C$, $C = \frac{-\omega_n^2}{s_b(s_a - s_b)}$

$$\Rightarrow C(s) = \frac{1}{s} - \frac{\omega_n}{2\sqrt{\xi^2-1}}\left(\frac{1}{s+s_a}\right)\frac{1}{s_a} + \frac{\omega_n}{2\sqrt{\xi^2-1}}\left[\frac{1}{s+s_b}\right]\frac{1}{s_b}$$

$$\Rightarrow C(s) = \frac{1}{s} - \frac{\omega_n}{2\sqrt{\xi^2-1}}\left[\frac{1}{(s+s_a)s_a} - \frac{1}{(s+s_b)s_b}\right].$$

$$\Rightarrow c(t) = 1 - \frac{\omega_n}{2\sqrt{\xi^2-1}}\left[\frac{e^{-s_a t}}{s_a} - \frac{e^{-s_b t}}{s_b}\right].$$



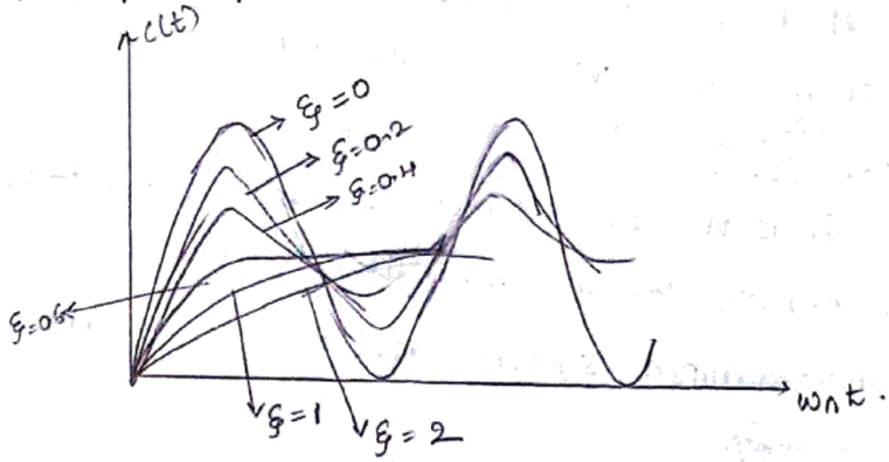
$$c(s) = \frac{\omega_n^2}{(s+s_a)(2\omega_n\sqrt{\xi^2-1})} + \frac{\omega_n^2}{(s+s_b)(-2\omega_n\sqrt{\xi^2-1})}$$

$$c(s) = \frac{\omega_n^2}{2\omega_n\sqrt{\xi^2-1}}\left[\frac{1}{s+s_a} - \frac{1}{s+s_b}\right]$$

Apply I.L

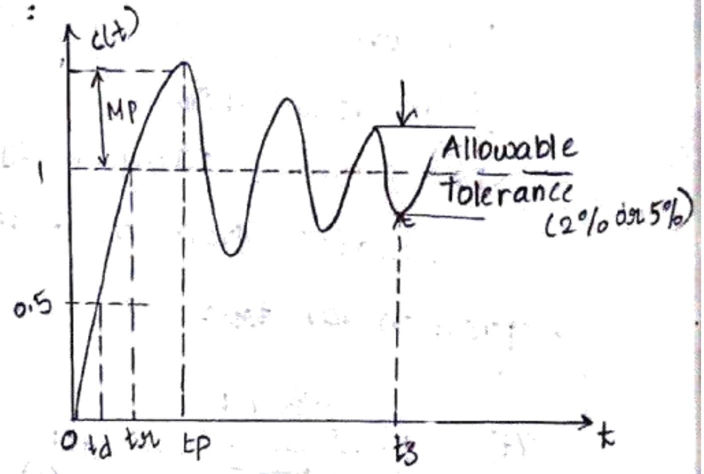
$$c(t) = \frac{\omega_n}{2\sqrt{\xi^2-1}}\left[e^{-s_a t} - e^{-s_b t}\right]$$

* Unit step Response of a second order system :



* Time domain Specifications :

Damped Oscillatory Response of a Second Order System



"The most practical standard is to start with the system at rest and so output and all time derivatives before $t=0$ will be zero".

The transient Response of practical control system often exhibits Damped oscillations before reaching steady state.

The transient Response characteristics of a control system to a unit step input is specified interms of following time domain specifications.

1. Delay time (td) ;
2. Rise time (tr)
3. Peak time (tp)
4. Maximum over shoot (Mp)
5. Settling time (ts)

1. Delay time (td) :

It is the time taken for the Response to reach 50% of the final value for the very first time.

2. Rise time (tr) :

It is the time taken for the response to rise from 0-100% for the very first time.

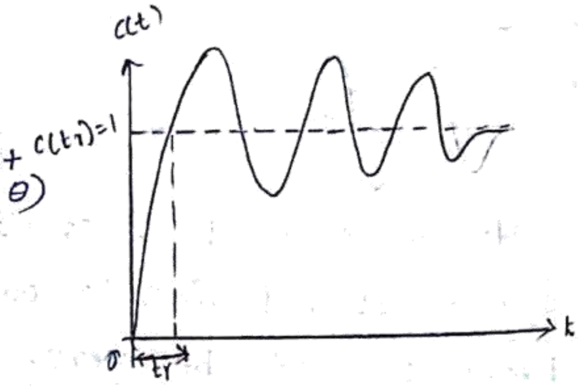
For under-damped system, rise time is calculated from 0-100%.

For over-damped system, rise time is calculated from 10% - 90%

For critically damped system, rise time is calculated from 5% - 95%.

Expression for tr :

$$c(t) = 1 - \frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}} \sin(\omega_d t + \theta)$$



At $t = t_r$, $c(t) = 1$

$$\Rightarrow 1 - \frac{e^{-\xi\omega_n t_r}}{\sqrt{1-\xi^2}} \sin(\omega_d t_r + \theta) = 1.$$

$$\Rightarrow \frac{e^{-\xi\omega_n t_r}}{\sqrt{1-\xi^2}} \sin(\omega_d t_r + \theta) = 0.$$

$\Rightarrow e^{-\xi\omega_n t_r} \neq 0$

$$\Rightarrow \frac{\sin(\omega_d t_r+\theta)}{\sqrt{1-\xi^2}} = 0.$$

$$\Rightarrow \sin(\omega_d t_r+\theta) = 0$$

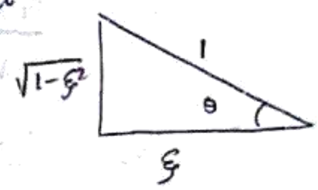
$$\Rightarrow \omega_d t_r + \theta = n\pi.$$

for $n=1$

$$\omega_d t_r + \theta = \pi$$

$$\boxed{t_r = \frac{\pi-\theta}{\omega_d}} = \pi-$$

$$\Rightarrow \boxed{t_r = \frac{\pi-\theta}{\omega_n\sqrt{1-\xi^2}}} \text{ rad}$$



$$\theta = \tan^{-1}\left[\frac{\sqrt{1-\xi^2}}{\xi}\right]$$

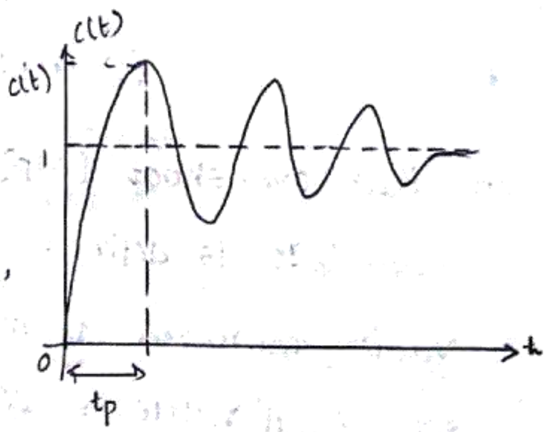
$$\Rightarrow t_r = \frac{\pi - \tan^{-1}\left(\frac{\sqrt{1-\xi^2}}{\xi}\right)}{\omega_n\sqrt{1-\xi^2}} \text{ sec.}$$

3. Peak Time (t_p) :

It is the time taken for the response to reach the peak value for the very first time.

(or)

It is the time taken for the response to reach the peak over shoot M_p.



$$c(t) = 1 - \frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}}\sin(\omega_d t+\theta)$$

Differentiating $c(t)$ w.r.t 't' and equating to zero.

$$\Rightarrow \frac{dc(t)}{dt} = 0.$$

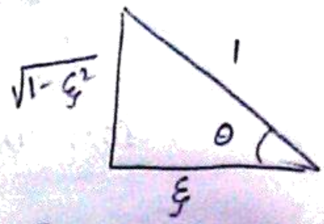
$$\Rightarrow \frac{-1}{\sqrt{1-\xi^2}}\left[-\xi\omega_n e^{-\xi\omega_n t}\sin(\omega_d t+\theta) + \frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}}\cos(\omega_d t+\theta)\omega_d\right] = 0$$

$$\Rightarrow \frac{-e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}}\left[-\xi\omega_n \sin(\omega_d t+\theta) + \omega_n\sqrt{1-\xi^2}\cos(\omega_d t+\theta)\right] = 0$$

$$\left[\because \omega_d = \omega_n\sqrt{1-\xi^2}\right]$$

$$\Rightarrow \frac{-\omega_n e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}}\left[-\xi \sin(\omega_d t+\theta) + \sqrt{1-\xi^2}\cos(\omega_d t+\theta)\right] = 0$$

From Δle $\cos\theta = \xi$, $\sqrt{1-\xi^2} = \sin\theta$



$$\Rightarrow \frac{-\omega_n e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}}\left[-\cos\theta \sin(\omega_d t+\theta) + \sin\theta\cos(\omega_d t+\theta)\right] = 0.$$

$$\Rightarrow \frac{\omega_n e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}}\left[\sin(\omega_d t+\theta)\cos\theta - \cos(\omega_d t+\theta)\sin\theta\right] = 0$$

$$\Rightarrow \frac{\omega_n e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}}\left[\sin(\omega_d t+\theta-\theta)\right] = 0.$$

$$\Rightarrow \frac{\omega_n e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}} \sin\omega_d t = 0$$

$\because e^{-\xi\omega_n t} \neq 0.$

At $t = t_p$, $\sin\omega_d t_p = 0$

$$\omega_d t_p = n\pi.$$

$$t_p = \frac{n\pi}{\omega_d}$$

for $n=1$, $t_p = \frac{\pi}{\omega_d} = \frac{\pi}{\omega_n\sqrt{1-\xi^2}}$

4. Peak over shoot [Mp] :

It is defined as ratio of maximum peak value measured from final value to final value

Let final value $= c(\infty)$,

maximum value $= c(t_p)$

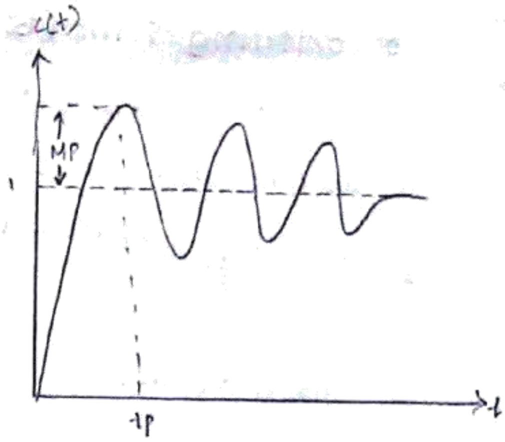
Peak overshoot $(M_p) = \frac{c(t_p) - c(\infty)}{c(\infty)}$

$$\% M_p = \frac{c(t_p) - c(\infty)}{c(\infty)} \times 100$$

expression for M_p :

$c(\infty)$ = final steady state value $=1$

$c(t_p)$ = peak response at $t=t_p$



$$c(t) = 1 - \frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}} \sin(\omega_d t + \theta)$$

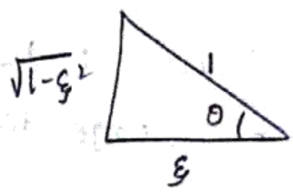
$$c(t_p) = 1 - \frac{e^{-\xi\omega_n t_p}}{\sqrt{1-\xi^2}} \sin(\omega_d t_p + \theta)$$

$$= 1 - \frac{e^{-\xi\omega_n \cdot \frac{\pi}{\omega_n\sqrt{1-\xi^2}}}}{\sqrt{1-\xi^2}} \sin\left[\omega_d \cdot \frac{\pi}{\omega_d} + \theta\right] \quad \left[\because t_p = \frac{\pi}{\omega_d} = \frac{\pi}{\omega_n\sqrt{1-\xi^2}}\right]$$

$$= 1 - \frac{e^{-\xi\frac{\pi}{\sqrt{1-\xi^2}}}}{\sqrt{1-\xi^2}} \sin(\pi + \theta).$$

$$= 1 - \frac{e^{-\xi\pi/\sqrt{1-\xi^2}}}{\sqrt{1-\xi^2}} (-\sin\theta)$$

$$= 1 + \frac{e^{-\xi\pi/\sqrt{1-\xi^2}}}{\sqrt{1-\xi^2}} \sin\theta$$



$$= 1 + \frac{e^{-\xi\pi/\sqrt{1-\xi^2}}}{\sqrt{1-\xi^2}} \cdot \sqrt{1-\xi^2}$$

$\left[\because \text{from } \Delta le \ \sin\theta = \sqrt{1-\xi^2}\right]$

$$c(t_p) = 1 + e^{-\xi\pi/\sqrt{1-\xi^2}}$$

Now we have

$$c(\infty) = 1.$$

$$\Rightarrow M_p = \frac{c(t_p) - c(\infty)}{c(\infty)} = \frac{1 + e^{-\xi\pi/\sqrt{1-\xi^2}} - 1}{1}$$

$$M_p = e^{-\xi\pi/\sqrt{1-\xi^2}}$$

$$\Rightarrow \boxed{\%\, M_p = e^{-\xi\pi/\sqrt{1-\xi^2}} \times 100}$$

5. Setteling Time (ts) :

It is defined as time taken by the response to reach and stay within a specified error. It is usually expressed as percentage of final value. The usual tolerable error is 2% (or) 5% of the final value.

Expression for ts :

The response of 2nd order system has 2 components, they are.

1. Decaying Exponential component. $\dfrac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}}$

2. Sinusoidal Component $\sin(\omega_d t + \theta)$

The Decaying Exponential term reduces oscillations produced by sinusoidal component. Hence the settling time is decided by exponential component. ts can be found out by equating Exponential component to percentage tolerance errors.

→ for 2% tolerance, at $t = ts$

$$\Rightarrow \frac{e^{-\xi\omega_n ts}}{\sqrt{1-\xi^2}} = 0.02$$

$$\Rightarrow e^{-\xi\omega_n ts} = 0.02\sqrt{1-\xi^2}$$

$$\Rightarrow \xi\omega_n ts = \ln\left[\frac{1}{0.02\sqrt{1-\xi^2}}\right]$$

$$\Rightarrow ts = \frac{1}{\xi\omega_n}\ln\left[\frac{1}{0.02\sqrt{1-\xi^2}}\right]$$

for small values of ξ, $\sqrt{1-\xi^2} = 1$.

$$\Rightarrow ts = \frac{1}{\xi\omega_n}\ln\left[\frac{1}{0.02}\right]$$

$$\Rightarrow t_s = \frac{4}{\xi\omega_n}$$

$$\Rightarrow t_s = 4T \quad \text{where } T = \frac{1}{\xi\omega_n}$$

T = time constant of the system.

→ for 5% tolerance

$$\frac{e^{-\xi\omega_n t_s}}{\sqrt{1-\xi^2}} = 0.05$$

$$\Rightarrow t_s = \frac{1}{\xi\omega_n} \ln\left(\frac{1}{0.05}\right)$$

$$\Rightarrow t_s = \frac{3}{\xi\omega_n}$$

$$\Rightarrow t_s = 3T \quad \text{where } T = \frac{1}{\xi\omega_n}$$

Generalised Expression of $t_s = \dfrac{\ln(\%\ \text{error})}{\xi\omega_n}$

* Expression of Time Domain Specifications :

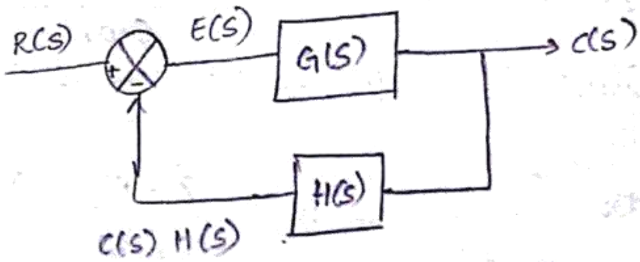
1.	Rise time - t_r	$t_r = \dfrac{\pi - \theta}{\omega_n\sqrt{1-\xi^2}} = \dfrac{\pi - \tan^{-1}\left(\frac{\sqrt{1-\xi^2}}{\xi}\right)}{\omega_n\sqrt{1-\xi^2}}$
2.	Peak time - t_p	$t_p = \dfrac{\pi}{\omega_d} = \dfrac{\pi}{\omega_n\sqrt{1-\xi^2}}$
3.	Peak overshoot - M_p	$M_p = e^{-\xi\pi/\sqrt{1-\xi^2}}$ $\% M_p = e^{-\xi\pi/\sqrt{1-\xi^2}} \times 100$
4.	Settling time - t_s	$t_s = \dfrac{\ln(\%\ \text{Error})}{\xi\omega_n}$, $T = \dfrac{1}{\xi\omega_n}$ for 2% tolerance $t_s = 4T$ for 5% tolerance $t_s = 3T$

STEADY-STATE RESPONSE :

Steady state Error :

The steady state error is the value of the error signal $e(t)$, when $t \to \infty$.

- Steady state error is a measure of system accuracy.
- These errors arise from nature of inputs, type of system and from non-linearity of system components.
- Steady-state performance of a stable control system is generally judged by its steady state error to step, ramp and parabolic inputs.



$$E(S) = R(S) - C(S)\,H(S) \longrightarrow (1)$$

$$C(S) = E(S) \cdot G(S) \longrightarrow (2)$$

subs. (2) in (1)

$$\Rightarrow E(S) = R(S) - E(S)\,G(S)\,H(S)$$

$$\Rightarrow E(S)\left[1 + G(S)H(S)\right] = R(S)$$

$$\Rightarrow E(S) = \frac{R(S)}{1 + G(S)\,H(S)}$$

Applying Inverse Laplace

$$\boxed{e(t) = L^{-1}(E(S)) = L^{-1}\left[\frac{R(S)}{1 + G(S)\,H(S)}\right]}$$

Let e_{ss} = steady state error.

It is defined as the value of $e(t)$ when $t \to \infty$. $\therefore$ $e_{ss} = \lim\limits_{t \to \infty} e(t)$

The final value theorem states that

If $F(s) = L[f(t)]$, then

$$\lim_{t \to \infty} f(t) = \lim_{s \to 0} s F(s)$$

using final value theorem, $e_{ss} = \lim_{s \to 0} s E(s)$

$$\Rightarrow \boxed{e_{ss} = \lim_{s \to 0} s \cdot \left[\frac{R(s)}{1+G(s)H(s)}\right]}$$

Type number of control systems :

Type number is specified for loop transfer function $G(s) \cdot H(s)$.

The number of poles of loop transfer function lying at the origin decides type number of the system.

In general, if N is the number of poles at the origin, then type number is N.

The loop transfer function can be expressed as ratio of two polynomials in s.

$$G(s)\, H(s) = K \frac{P(s)}{Q(s)} = K \frac{(s+z_1)(s+z_2)(s+z_3)+ \cdots}{s^N (s+P_1)(s+P_2)(s+P_3) \cdots}$$

where z_1, z_2, z_3 are zeroes of the transfer function

P_1, P_2, P_3 are poles of the transfer function

K is a constant

N is number of poles at origin.

The value of N in the denominator polynomial decides the type of number of the system.

If $N = 0$, then it 0- type system

$N = 1$, type-1 system

$N = 2$, type-2 system .....

* Steady state Error when input is unit step signal :

$$e_{ss} = \lim_{s \to 0} \frac{sR(s)}{1+G(s)H(s)}.$$

$r(t) = u(t)$

$R(s) = \frac{1}{s}.$

$$\Rightarrow e_{ss} = \underset{s\to 0}{Lt}\ s\left[\frac{1}{s}\right]\left[\frac{1}{1+G(s)H(s)}\right]$$

$$\Rightarrow e_{ss} = \underset{s\to 0}{Lt}\ \frac{1}{1+G(s)H(s)}$$

$$\Rightarrow e_{ss} = \frac{1}{1+\underset{s\to 0}{Lt}\ G(s)H(s)} = \frac{1}{1+K_p}$$

where K_p = positional Error constant.

$$e_{ss} = \frac{1}{1+K_p}. \quad \text{where } K_p = \underset{s\to 0}{Lt}\ G(s)H(s)$$

Type - 0 system :

$$K_p = \underset{s\to 0}{Lt}\ G(s)H(s)$$

$$\Rightarrow K_p = \underset{s\to 0}{Lt}\ K\frac{(s+z_1)(s+z_2)(s+z_3)\cdots}{s^0 s^N (s+P_1)(s+P_2)(s+P_3)\cdots}$$

$$\Rightarrow K_p = K\cdot\frac{z_1 z_2 z_3\cdots}{P_1 P_2 P_3\cdots} = \text{constant}.$$

$$\Rightarrow e_{ss} = \frac{1}{1+K_p} - \text{constant}.$$

Type - 1 system :

$$K_p = \underset{s\to 0}{Lt}\ G(s)H(s)$$

$$= \underset{s\to 0}{Lt}\ \frac{K(s+z_1)(s+z_2)(s+z_3)\cdots}{s^1(s+P_1)(s+P_2)(s+P_3)\cdots}$$

$$K_p = \infty$$

$$\therefore e_{ss} = \frac{1}{1+K_p} = \frac{1}{\infty} = 0$$

$$\therefore e_{ss} = 0$$

In systems, with type number -1 & above for unit step input, the value of K_p is infinite and so

$\boxed{e_{ss} = 0}$

* Steady state Error for Ramp Input :

$$e_{ss} = \lim_{s \to 0} \frac{sR(s)}{1+G(s)H(s)}$$

$$R(s) = \frac{1}{s^2}$$

$$\Rightarrow e_{ss} = \lim_{s \to 0} s\left[\frac{1}{s^2}\right]\left[\frac{1}{1+G(s)H(s)}\right]$$

$$\Rightarrow e_{ss} = \lim_{s \to 0} \frac{1}{s[1+G(s)H(s)]}$$

$$\Rightarrow e_{ss} = \frac{1}{0 + \lim_{s \to 0} G(s)H(s)}$$

$$\Rightarrow e_{ss} = \frac{1}{\lim_{s \to 0} sG(s)H(s)} = \frac{1}{K_v}$$

where $K_v = \lim_{s \to 0} sG(s)H(s)$

K_v = velocity error constant.

$$\therefore e_{ss} = \frac{1}{K_v}$$

Type-0 system :

$$K_v = \lim_{s \to 0} sG(s)H(s)$$

$$\Rightarrow K_v = \lim_{s \to 0} \frac{Ks \cdot (s+z_1)(s+z_2)(s+z_3)\cdots}{s^0 (s+P_1)(s+P_2)(s+P_3)\cdots}$$

$$\Rightarrow K_v = 0$$

$$\Rightarrow e_{ss} = \frac{1}{K_v} = \frac{1}{0} = \infty$$

$$e_{ss} = \infty$$

Type-1 system :

$$K_v = \lim_{s \to 0} sG(s)H(s)$$

$$\Rightarrow K_v = \lim_{s \to 0} \frac{Ks\,(s+z_1)(s+z_2)(s+z_3)\cdots}{s(s+P_1)(s+P_2)(s+P_3)\cdots}$$

$$\Rightarrow K_v = K\frac{z_1 z_2 z_3 \cdots}{P_1 P_2 P_3 \cdots} = \text{constant}$$

$$\Rightarrow e_{ss} = \frac{1}{K_v} = \text{constant}.$$

Type-2 system :

$$K_v = \lim_{s \to 0} s G(s) H(s)$$

$$K_v = \lim_{s \to 0} \frac{K s (s+z_1)(s+z_2) \cdots}{s^2 (s+P_1)(s+P_2) \cdots}$$

$$K_v = \lim \infty$$

$$\Rightarrow e_{ss} = \frac{1}{K_v} = \frac{1}{\infty} = 0$$

$$\Rightarrow e_{ss} = 0$$

In system type-2 and above for ramp input, the value of K_v is infinite $\Rightarrow e_{ss} = 0$.

* Steady state error for Parabolic Input signal :

$$e_{ss} = \lim_{s \to 0} \frac{s R(s)}{1 + G(s) H(s)}$$

$$R(s) = \frac{1}{s^3}$$

$$\Rightarrow e_{ss} = \lim_{s \to 0} \frac{1}{s^2 (1 + G(s) H(s))}$$

$$\Rightarrow e_{ss} = \frac{1}{\lim_{s \to 0} s^2 G(s) H(s)} = \frac{1}{K_a}$$

where $K_a = \lim_{s \to 0} s^2 G(s) H(s)$

K_a = acceleration error constant.

Type-0 system :

$$K_a = \lim_{s \to 0} s^2 G(s) H(s).$$

$$\Rightarrow K_a = \lim_{s \to 0} \frac{K s^2 (s+z_1)(s+z_2) \cdots}{s^0 (s+P_1)(s+P_2) \cdots}$$

$$\Rightarrow K_a = 0$$

$$\Rightarrow e_{ss} = \frac{1}{K_a} = \infty$$

$$e_{ss} = \infty$$

Type-1 system :

$$K_a = \lim_{s\to 0} Ks^2\left[\frac{(s+z_1)(s+z_2)\ldots}{s(s+P_1)(s+P_2)\ldots}\right]$$

$$K_a = 0$$

$\Rightarrow e_{ss} = \infty$

Type-2 system :

$$K_a = \lim_{s\to 0} Ks^2\left[\frac{(s+z_1)(s+z_2)\ldots}{s^2(s+P_1)(s+P_2)\ldots}\right]$$

$$\Rightarrow K_a = K\frac{z_1 z_2 z_3}{P_1 P_2 P_3} = \text{constant}$$

$$\Rightarrow e_{ss} = \frac{1}{K_a} = \text{constant.}$$

Type-3 system :

$$K_a = \lim_{s\to 0} \frac{Ks^2(s+z_1)(s+z_2)\ldots}{s^3(s+P_1)(s+P_2)\ldots}$$

$$\Rightarrow K_a = \infty$$

$$\Rightarrow e_{ss} = \frac{1}{K_a} = 0.$$

In system type-3 and above for parabolic input signal, the value of K_a is $\infty \rightarrow e_{ss} = 0$

* Summary Table of steady state Errors for various inputs :

Type Number	Ess when Input signal is		
	step	Ramp	parabolic
0	$\frac{1}{1+K_p}$	∞	∞
1.	0	$\frac{1}{K_v}$	∞
2	0	0	$\frac{1}{K_a}$
3	0	0	0

* Error constants for various inputs :

Type Number	Error constant when i/p signal is		
	step	Ramp	Parabolic
0	Kp	0	0
1	∞	Kv	0
2	∞	∞	Ka
3	∞	∞	∞

* Static Error coefficients :

When a control system is excited with a standard input signal, the e_{ss} (steady state error) may be constant, 0, ∞.

- Type-0 system will have constant ess when input is step signal. That constant is called positional error constant (Kp).

$$K_p = \lim_{s \to 0} G(s) H(s)$$

- Type-1 system will have constant ess when input is ramp signal. That constant is called velocity error constant (Kv)

$$K_v = \lim_{s \to 0} sG(s) H(s)$$

- Type-2 system will have constant ess when input is Parabolic signal. That constant is called acceleration error constant (Ka).

$$K_a = \lim_{s \to 0} s^2 G(s) H(s).$$

Kp, Kv and Ka are static Error constants.

* Generalized Error Coefficients :

The drawback of static error coefficients is that it does not show the variation of error with time and input should be a standard input.

- Generalised Error coefficients gives ess as a function of time and can be found for any type of input.

$$E(s) = \frac{R(s)}{1+G(s)H(s)}$$

$$\Rightarrow E(s) = \frac{1}{1+G(s)H(s)} \cdot R(s) = F(s) \cdot R(s)$$

where $F(s) = \frac{1}{1+G(s)H(s)}$

Let $e(t) = L^{-1}[E(s)]$

$f(t) = L^{-1}[F(s)]$

$r(t) = L^{-1}[R(s)]$

The convolution Theorem states that the Inverse Laplace Transform of the product of two s-domain functions is equal to convolution of their time domain functions.

$$L^{-1}[F(s)R(s)] = \int_{-\infty}^{\infty} f(T)r(t-T)dT$$

$$\therefore e(t) = \int_{-\infty}^{\infty} f(T)r(t-T)dT$$

$$\Rightarrow e(t) = \int_{0}^{\infty} f(T)r(t-T)dT$$

using Taylor series expansion, the signal $r(t-T)$ can be expressed as

$$r(t-T) = r(t) - T\dot{r}(t) + \frac{T^2}{2!}\ddot{r}(t) - \frac{T^3}{3!}\dddot{r}(t) + \dots (-1)^n \frac{T^n}{n!} r^{n}(t)$$

$$\Rightarrow e(t) = \int_{0}^{\infty} f(T)\left[r(t) - T\dot{r}(t) + \frac{T^2}{2!}\ddot{r}(t) - \dots (-1)^n \frac{T^n}{n!} r^{n}(t)\right] dT$$

$$\Rightarrow e(t) = \int_{0}^{\infty}\left[f(T)r(t) - f(T)T\dot{r}(t) + \frac{T^2}{2!}f(T)\ddot{r}(t) - \dots (-1)^n \frac{T^n}{n!} f(T)r^{n}(t)\right] dT$$

$$e(t) = \int_0^\infty f(T)\, r(t)\, dT - \int_0^\infty f(T)\, T\, \dot{r}(t)\, dT + \int_0^\infty f(T) \frac{T^2}{2!} \ddot{r}(t) - \cdots$$

$$\int_0^\infty (-1)^n f(T) \frac{T^n}{n!} \overset{n}{r}(t)\, dT$$

Since $r(t), \dot{r}(t), \ddot{r}(t), \ldots$ are constants, the integration is done w.r.t T. The error signal can be written as

$$e(t) = r(t) \int_0^\infty f(T)\, dT - \dot{r}(t) \int_0^\infty f(T)\, T\, dT + \frac{\ddot{r}(t)}{2!} \int_0^\infty f(T)\, T^2\, dT - \cdots$$

$$(-1)^n \frac{\overset{n}{r}(t)}{n!} \int_0^\infty f(T)\, T^n\, dT$$

Let $\int_0^\infty f(T)\, dT = C_0$, $\int_0^\infty -T f(T)\, dT = C_1, \ldots (-1)^n \int_0^\infty T^n f(T) = C_n$

$$\therefore e(t) = r(t) C_0 + \dot{r}(t) C_1 + \frac{\ddot{r}(t)}{2!} C_2 - \cdots (-1)^n \frac{\overset{n}{r}(t)}{n!} C_n$$

The coefficients $C_0, C_1, \ldots C_n$ are called as generalised error coefficients (or) dynamic error coefficients.

The e_{ss} is obtained by taking limit $t \to \infty$ on $e(t)$

$$\Rightarrow \boxed{e_{ss} = \lim_{t \to \infty} e(t)}$$

Evaluation of Generalised Error Coefficients :

The generalized error coefficient is given by

$$C_n = (-1)^n \int_0^t T^n f(T)\, dT \;; \quad \text{where} \quad F(s) = \frac{1}{1 + G(s)H(s)}$$

We know that $L\{F(T)\} = F(s)$, hence by the definition of L.T

$$F(s) = \int_0^t f(T)\, e^{-sT}\, dT \quad \text{— (1)}$$

On taking $\lim_{s \to 0}$ on both sides of (1)

$$\lim_{s \to 0} F(s) = \lim_{s \to 0} \int_0^t f(T)\, e^{-sT}\, dT$$

$$= \int_0^t f(T) \lim_{s \to 0} e^{-sT}\, dT$$

$$= \int_0^t f(T)\, dT = C_0$$

$$c_0 = \underset{s \to 0}{Lt} F(s) \text{ --- } ②$$

On diff eqn ② w.r.t s.

$$\frac{d}{ds} F(s) = \frac{d}{ds} \int_0^t f(T) e^{-sT} dT$$

$$= \int_0^t f(T) \frac{d}{ds} e^{-sT} dT$$

$$= \int_0^t f(T)(-T) e^{-sT} dT$$

$$= -\int_0^t T f(T) e^{-sT} dT \text{ --- } ③$$

On taking $\underset{s \to 0}{Lt}$ on both sides of ③

$$\underset{s \to 0}{Lt} \frac{d}{ds} F(s) = \underset{s \to 0}{Lt} - \int_0^t T f(T) e^{-sT} dT$$

$$= -\int_0^t T f(T) \underset{s \to 0}{Lt} e^{-sT} dT$$

$$= -\int_0^t T f(T) dT = c_1$$

$$c_1 = \underset{s \to 0}{Lt} \frac{d}{ds} F(s) \text{ --- } ④$$

On diff eqn ② on b.s w.r.t. s

$$\frac{d}{ds}\left[\frac{d}{ds} F(s)\right] = \frac{d}{ds}\left[-\int_0^t T f(T) e^{-sT} dT\right]$$

$$\frac{d^2}{ds^2} F(s) = \left[-\int_0^t T f(T) \frac{d}{ds}(e^{-sT}) dT\right]$$

$$= -\int_0^t T f(T)(-T) e^{-sT} dT$$

$$\frac{d^2 F(s)}{ds^2} = \int_0^t T^2 f(T) e^{-sT} dT \text{ --- } ⑤$$

Apply $s \to 0$ on b.s of ⑤.

$$\underset{s \to 0}{Lt} \frac{d^2}{ds^2} F(s) = \underset{s \to 0}{Lt} \int_0^t T^2 f(T) e^{-sT} dT$$

$$= \int_0^t T^2 f(T) \underset{s \to 0}{Lt} e^{-sT} dT$$

$$= \int_0^t \tau^2 f(\tau)\, d\tau = C_2$$

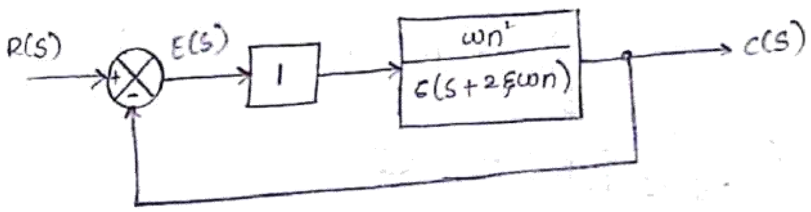
$$C_2 = \lim_{s \to 0} \frac{d^2}{ds^2} F(s) \quad \text{--- (6)}$$

similarly It can be shown that

$$C_n = \lim_{s \to 0} \frac{d^n}{ds^n} F(s)$$

* **Introduction to PID controllers :**

A controller is a device which when introduced in feedback or forward path of a system, which controls steady-state and transient response as per the requirement. In most of the practical systems, controller input is proportional to error generator. such systems are called proportional to error mechanisms or P- type.



$$G(S)\,H(S) = \frac{\omega_n^2}{S(S+2\xi\omega_n)}$$

$$\frac{C(S)}{R(S)} = \frac{G(S)}{1+G(S)\,H(S)} = \frac{\frac{\omega_n^2}{S(S+2\xi\omega_n)}}{1+\frac{\omega_n^2}{S(S+2\xi\omega_n)}}$$

$$\frac{C(S)}{R(S)} = \frac{\omega_n^2}{S^2+2\xi\omega_n S+\omega_n^2}$$

For this system, damping ratio is ξ and natural frequency ω_n . and steady state error K_p

$$K_p = \lim_{S \to 0} G(S)\,H(S)$$

$$= \lim_{S \to 0} \frac{\omega_n^2}{S(S+2\xi\omega_n)} = \infty$$

$$\Rightarrow \boxed{K_p = \infty}$$

$$K_V = \lim_{s\to 0} sG(s)H(s)$$

$$= \lim_{s\to 0} s\left[\frac{\omega_n^2}{s(s+2\xi\omega_n)}\right] = \frac{\omega_n^2}{2\xi\omega_n} = \frac{\omega_n}{2\xi}$$

$$\boxed{K_V = \frac{\omega_n}{2\xi}}$$

$$K_a = \lim_{s\to 0} s^2G(s)H(s) = \lim_{s\to 0} \frac{\omega_n^2 \cdot s^2}{s(s+2\xi\omega_n)} = 0$$

$$\Rightarrow \boxed{K_a = 0}$$

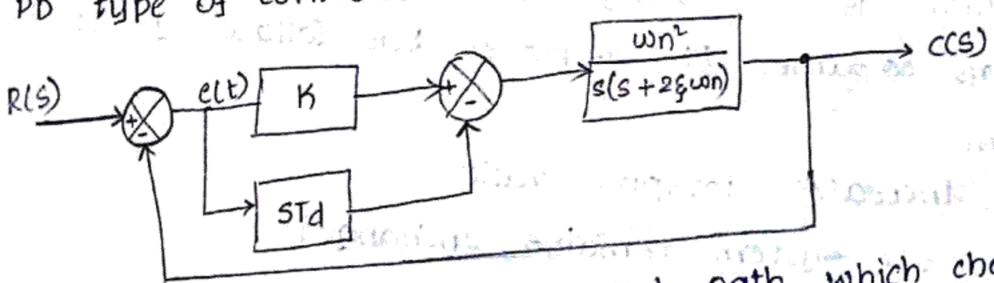
If transient response is to be improved, damping ratio must be changed. By increasing K_V, ess can be reduced. But due to high gain settling time and peak overshoot increases. This may lead to instability of the system.

- In general good time response demands
1. less settling time
2. less overshoot
3. less rise time
4. smallest steady state error.

So, as a compromise is made to keep ess and overshoot between acceptable limits by providing following different types of controllers.

1. PD - Proportional + Derivative action
2. PI - Proportional + Integral action
3. PID - Proportional + Integral + Derivative action.

* PD type of controller :



A controller in the forward path which changes the controller input to the proportional + Derivative of error signal is called PD controller.

Input to the controller $= K e(t) + T_d \frac{de(t)}{dt}$

Taking Laplace $= KE(s) + sT_dE(s)$

$= E(s)[K + sT_d]$

Transfer function $= K + ST_d$

Assuming $K = 1$

$$G(S) = \frac{(1+ST_d)\omega_n^2}{S(S+2\xi\omega_n)}$$

$$\frac{C(S)}{R(S)} = \frac{(1+ST_d)\omega_n^2}{S^2 + S[2\xi\omega_n + \omega_n^2 T_d] + \omega_n^2}$$

comparing denominator with standard form ω_n is same as P-type controller and

$$2\xi\omega_n = 2\xi\omega_n + \omega_n^2 T_d$$

$$\xi = \frac{2\xi\omega_n + \omega_n^2 T_d}{2\omega_n}$$

$$\xi = \xi + \frac{\omega_n T_d}{2}$$

Because of this controller, damping ratio increases by a factor $\frac{\omega_n T_d}{2}$

$$K_p = \lim_{S\to 0} \frac{(1+ST_d)\omega_n^2}{S(S+2\xi\omega_n)} = \infty$$

$$K_v = \lim_{S\to 0} \frac{S(1+ST_d)\omega_n^2}{S(S+2\xi\omega_n)}$$

$$= \frac{\omega_n^2}{2\xi\omega_n} = \frac{\omega_n}{2\xi}$$

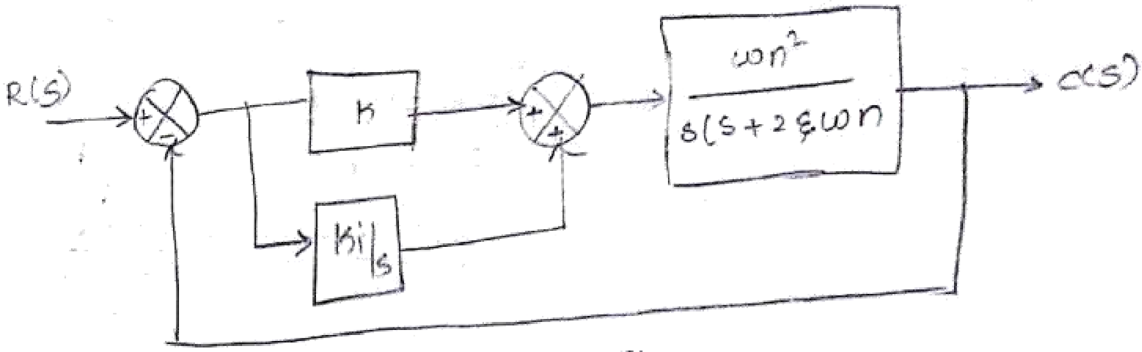
$$K_v = \frac{\omega_n}{2\xi}$$

As there is no change in coefficients, error also will remains ~~as~~ same. PD controller has following effects on system

1. It increases damping ratio.
2. ω_n for system remains unchanged.
3. Type of system remains unchanged.
4. It reduces peak overshoot.
5. It reduces settling time.
6. Steady state error remains unchanged.

Transient part is include.

* PI type of controller :



Transfer function = K + Ki/s.

Assume K = 1.

$$G(s) = \frac{\omega_n^2(1+Ki/s)}{s(s+2\xi\omega_n)}$$

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2(1+Ki/s)}{s(s+2\xi\omega_n)+\omega_n^2(1+Ki/s)}$$

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2(1+Ki/s)}{s^2+s(2\xi\omega_n)+\frac{Ki\omega_n^2}{s}+\omega_n^2}$$

$$\frac{C(s)}{R(s)} = \frac{s(\omega_n^2(s+Ki))}{s^3+s^2(2\xi\omega_n)+s\omega_n^2+Ki\omega_n^2}$$

Since the order is increased by one, system is relatively less stable and Ki must be designed in such a way that system will remain in stable condition.

second order system is always stable.

Transient Response gets effected badly if controller is not designed properly.

$$K_p = \lim_{s\to0} G(s)\cdot H(s) = \lim_{s\to0} \frac{\omega_n^2(1+\frac{Ki}{s})}{s(s+2\xi\omega_n} = \infty$$

$$K_v = \lim_{s\to0} sG(s)H(s) = \lim_{s\to0} \frac{s(\omega_n^2+\frac{Ki}{s})}{s(s+2\xi\omega_n)} = \infty$$

$$K_a = \lim_{s \to 0} s^2 G(s)H(s) = \lim_{s \to 0} \frac{s^2 (1 + K_i/s)\omega_n^2}{s(s + 2\xi\omega_n)}$$

$$= \lim_{s \to 0} \frac{(s^2 + sK_i)\omega_n^2}{s^3 + 2s^2\xi\omega_n}$$

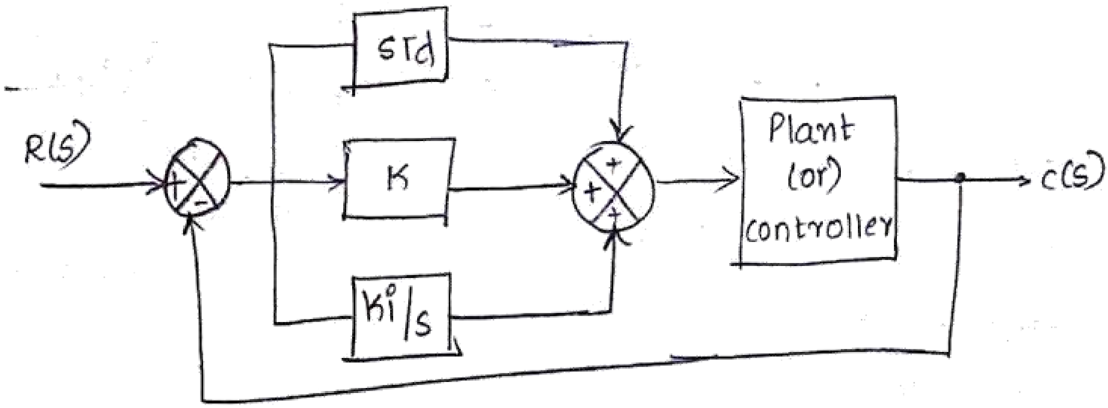
$$= \lim_{s \to 0} \frac{\omega_n^2 K_i}{2\omega_n \xi} = \frac{K_i\omega_n}{2\xi}$$

$$e_{ss} = \frac{1}{K_a} = \frac{2\xi}{K_i\omega_n}$$

Effects of PI controller :

1. It increases order and type of the system.
2. e_{ss} reduces tremendously for same type of inputs i.e., in general, this controller improves steady state part rather than transient part.
3. System is relatively less stable.

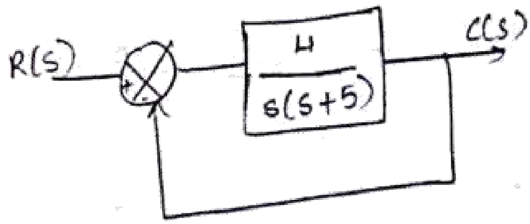
* PID Controller :



Since PD improves transient Response and PI improves steady state Response, the combination of both may be used to improve overall response of the system.

1. Obtain the response of unity feedback system whose open loop transfer function is $G(s) = \frac{4}{s(s+5)}$ and when the input is unit step.

Sol. $$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)H(s)}$$



$$= \frac{\frac{4}{s(s+5)}}{1+\frac{4}{s(s+5)}} \quad [\because H(s)=1]$$

$$= \frac{4}{s(s+5)+4} = \frac{4}{s^2+5s+4} = \frac{4}{(s+4)(s+1)}$$

Given $r(t) = u(t) \Rightarrow R(s) = 1/s$

$$\Rightarrow C(s) = \frac{4}{(s+4)(s+1)} \cdot R(s) = \frac{4}{s(s+4)(s+1)}$$

$$C(s) = \frac{A}{s} + \frac{B}{s+4} + \frac{C}{s+1}$$

$4 = A(s+4)(s+1) + B(s)(s+1) + Cs(s+4)$

$$C(s) = \frac{1}{s} + \frac{1}{3(s+4)} - \frac{4}{3(s+1)}$$

$4 = A(4)$

$\boxed{A = 1}$ $s = -4$

$4 = B(-4)(-3)$

$$c(t) = u(t)\left[1 + \frac{1}{3}e^{-4t} - \frac{4}{3}e^{-t}\right]$$

$B = \frac{1}{3}$

$s = -1$

$4 = -C(+3)$

$C = \frac{-4}{3}$

2. Measurements are conducted on a servo mechanism to show the system response $c(t) = 1 + 0.2e^{-60t} - 1.2e^{-10t}$, when input is unit step. Obtain the expression for closed loop transfer function. Determine undamped natural frequency and damping ratio.

Sol:- Given $c(t) = 1 + 0.2e^{-60t} - 1.2e^{-10t}$

$$C(s) = \frac{1}{s} + 0.2\frac{1}{s+60} - 1.2\frac{1}{s+10}.$$

Given $r(t) = u(t) \Rightarrow R(s) = \frac{1}{s}$.

$$\Rightarrow \frac{C(s)}{\cancel{R}(s)} = \cancel{s}\left[\frac{1}{s} + \frac{0.2}{s+60} - \frac{1.2}{s+10}\right]$$

$$\Rightarrow C(s) = \frac{(s+60)(s+10) + 0.2s(s+10) - 1.2s(s+60)}{s(s+60)(s+10)}$$

$$= \frac{\cancel{s^2} + \cancel{70s} + 600 + \cancel{0.2s^2} + \cancel{2s} - \cancel{1.2s^2} - \cancel{72s}}{s(s+60)(s+10)}$$

$$= \frac{600}{s(s+60)(s+10)}$$

$$\Rightarrow C.S = T.F \times R(s) \Rightarrow T.F = s(C(s))$$

$$\Rightarrow T.F = \frac{600}{s(s+60)(s+10)} \times s = \frac{600}{s^2+70s+600}$$

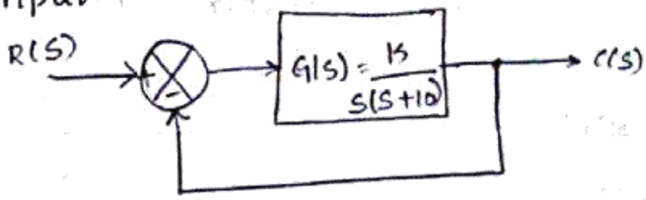
comparing with standard T.F $\frac{\omega_n^2}{s^2+2\xi\omega_n s+\omega_n^2}$

$$\Rightarrow \omega_n^2 = 600 \qquad 2\xi\omega_n = 70.$$

$$\omega_n = \sqrt{600} \qquad \xi = \frac{70}{2\sqrt{600}} = 1.428$$

$$\omega_n = 24.49 \text{ rad/s}$$

3. The unity feedback system is characterized by a open loop transfer function $G(s) = \frac{K}{s(s+10)}$. Determine the gain K, so that the system will have damping ratio of 0.5 for this value of K. Determine T_s, peak over shoot M_p and t_p. for a unit step Input.



Sol:-

$$\frac{C(s)}{R(s)} = \frac{G(s)}{G(s)H(s)+1}$$

$$= \frac{\frac{K}{s(s+10)}}{1+\frac{K}{s(s+10)}} = \frac{K}{s^2+10s+K}$$

comparing with the standard transfer function $\frac{\omega_n^2}{s^2+2\xi\omega_n s+\omega_n^2}$

$\Rightarrow \omega_n^2 = K$; $2\xi\omega_n = 10$

$\omega_n = \sqrt{K}$; $\xi\omega_n = 5$.

$\Rightarrow \omega_n = 10 = \sqrt{K}$; $\Rightarrow \xi = 0.5$ (given)

$\Rightarrow K = 100$; $\Rightarrow \omega_n = \frac{5}{0.5} = 10$ rad/sec

$$t_p = \frac{\pi}{\omega_n\sqrt{1-\xi^2}} = \frac{\pi}{10\sqrt{1-(0.5)^2}} = 0.362\ s.$$

$$M_p = e^{-\xi\pi/\sqrt{1-\xi^2}} = 0.2416 = 24\% \ 16\%$$

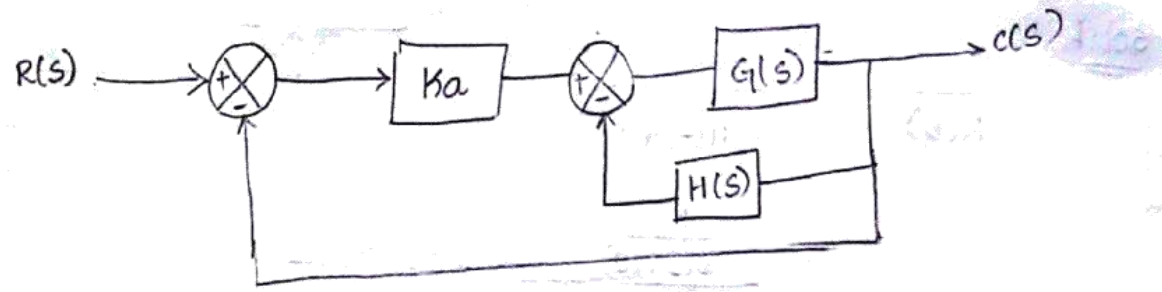
t_s = for 2% tolerance $= \frac{4}{\xi\omega_n} = 0.8\ s$

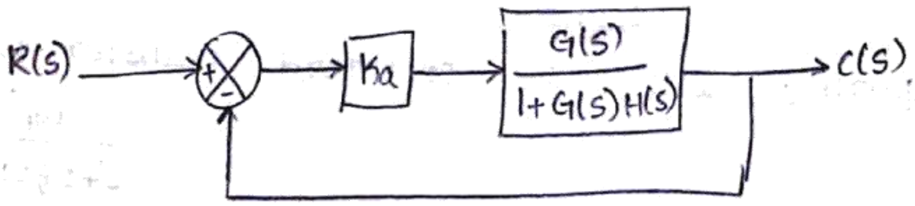
t_s = for 5% tolerance $= \frac{3}{\xi\omega_n} = 0.6\ s.$

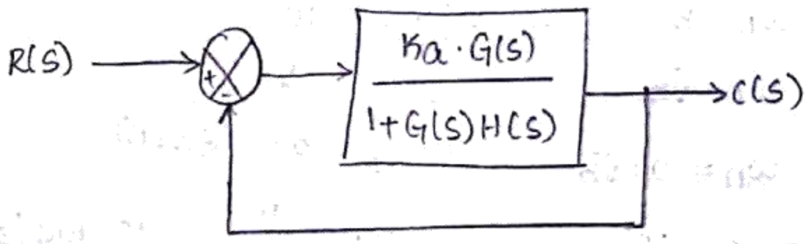
(or) $t_s = \frac{\ln\left[\frac{1}{\%\ \text{Error}}\right]}{\xi\omega_n}$

4. A unity feedback control system has an amplifier with gain $K_a = 10$ and gain ratio $G(s) = \frac{1}{s(s+2)}$ in the feed forward path. A Derivative feedback $H(s) = sK_o$ is introduced as a minor loop around $G(s)$. Determine the derivative feedback constant K_o so that the system damping ratio is 0.6.

Sol:-







$$\frac{K_a G(s)}{1+G(s)H(s)} = 10\left[\frac{\frac{1}{s(s+2)}}{1+\frac{1}{s(s+2)}sK_o}\right]$$

$$= \frac{10}{s}\left[\frac{1}{s+2+K_o}\right]$$

$$= \frac{10}{s(s+2+K_o)}$$

$$\frac{C(s)}{R(s)} = \frac{\frac{10}{s(s+2+K_o)}}{1+\frac{10}{(s+2+K_o)s}} = \frac{10}{s(s+2+K_o)+10} = \frac{10}{s^2+(2+K_o)s+10}$$

$$\Rightarrow \frac{C(s)}{R(s)} = \frac{10}{s^2+s(2+K_0)s+10}$$

comparing with $\frac{\omega_n^2}{s^2+2\xi\omega_n s+\omega_n^2}$.

$2\xi\omega_n = 2+K_0$ $\qquad \omega_n = \sqrt{10}$, $\xi = 0.6$

$\Rightarrow 2(0.6)(\sqrt{10}) - 2 = K_0$

$\Rightarrow K_0 = 1.79$

5. For a unity feedback system, the open loop transfer function $G(s) = \frac{10(s+2)}{s^2(s+1)}$. Find K_p, K_v, K_a. Also find e_{ss} when input is $R(s)$ where $R(s) = \frac{3}{s} - \frac{2}{s^2} + \frac{1}{3s^3}$.

Sol:-

$$K_p = \lim_{s\to 0} G(s)H(s) = \lim_{s\to 0} \frac{10(s+2)}{s^2(s+1)} = \infty$$

$$K_v = \lim_{s\to 0} s\,G(s)H(s) = \lim_{s\to 0} \frac{10(s+2)s}{s^2(s+1)} = \infty$$

$$K_a = \lim_{s\to 0} s^2 G(s)H(s) = \lim_{s\to 0} \frac{s^2\,10(s+2)}{s^2(s+1)} = \frac{10\left[2\left(\frac{s}{2}+1\right)\right]}{[s+1]}$$

$K_a = 20.$

$e_{ss1} = \frac{1}{K_p} = \frac{1}{\infty} = 0$,

$e_{ss2} = \frac{1}{K_v} = \frac{1}{\infty} = 0$, $\quad e_{ss3} = \frac{1}{K_a} = \frac{1}{20} = 0.05$

$$e_{ss} = \lim_{s\to 0} s \cdot \frac{R(s)}{1+G(s)H(s)}$$

$$= \lim_{s\to 0} \frac{s\left(\frac{3}{s} - \frac{2}{s^2} + \frac{1}{3s^3}\right)}{1+\frac{10(s+2)}{s^2(s+1)}} = \frac{3-\frac{2}{s}+\frac{1}{3s^2}}{\frac{s^2(s+1)+10(s+2)}{s^2(s+1)}}$$

$$= \lim_{s\to 0} \frac{9s^2-6s+1}{3\cancel{s^2}} \times \frac{\cancel{s^2}(s+1)}{s^2(s+1)+10(s+2)}$$

$$= \lim_{s \to 0} \frac{(9s^2 - 6s + 1)(s+1)}{3(s^2(s+1) + 10(s+2))}$$

$$= \lim_{s \to 0} \frac{(9s^2 - 6s + 1)(s+1)}{3[s^3 + s^2 + 10s + 20]}$$

$$e_{ss} = \frac{1}{60}$$

6 For a system $G(s)H(s) = \frac{K}{s^2(s+2)(s+3)}$. Find the value of K to limit steady state error to 10 when input to the system is $1 + 10t + \frac{40t^2}{2}$.

Sol:-

$$K_p = \lim_{s \to 0} G(s)H(s) = \lim_{s \to 0} \frac{K}{s^2(s+2)(s+3)} = \infty \Rightarrow e_{ss_1} = \frac{1}{K_p} = 0$$

$$K_v = \lim_{s \to 0} sG(s)H(s) = \lim_{s \to 0} \frac{Ks}{s^2(s+2)(s+3)} = \infty \Rightarrow e_{ss_2} = \frac{1}{K_v} = 0$$

$$K_a = \lim_{s \to 0} s^2G(s)H(s) \cdot \lim_{s \to 0} \frac{s^2 K}{s^2(s+2)(s+3)} = \frac{K}{6} \Rightarrow e_{ss} = \frac{1}{K_a} = \frac{6}{K}$$

we know $e_{ss} = e_{ss_1} + e_{ss_2} + e_{ss_3}$

From $r(t) = 1 + 10t + \frac{40t^2}{2}$, $A_1 = 1$, $A_2 = 10$, $A_3 = \frac{40}{2} = 20$.

$$\Rightarrow e_{ss} = e_{ss_1} + e_{ss_2} + e_{ss_3} = \frac{A_1}{1+K_p} + \frac{A_2}{K_v} + \frac{A_3}{K_a}$$

$$\Rightarrow 10 = \frac{6}{40K}[40]$$

$$\Rightarrow K = 24.$$

7. The loop transfer function of a feedback control system is given by $G(s)H(s) = \frac{100}{s^2(s+4)(s+12)}$. Determine the static error coefficients. Also determine e_{ss} for the input $r(t) = 2t^2 + 5t + 10$.

Sol:-

$$K_p = \lim_{s \to 0} G(s)H(s) = \lim_{s \to 0} \frac{100}{s^2(s+4)(s+12)} = \infty$$

$$K_v = \lim_{s\to 0} s\,G(s)H(s) = \infty$$

$$K_a = \lim_{s\to 0} s^2 G(s)H(s) = \frac{s^2 100}{s^2(s+4)(s+12)} = \frac{100}{48}$$

$$e_{ss} = e_{ss1} + e_{ss2} + e_{ss}.$$

$$e_{ss} = \frac{A_1}{1+K_p} + \frac{A_2}{K_v} + \frac{A_3}{K_a}$$ where $A_1 = 10$, $A_2 = 5$, $A_3 = 4$. from $r(t)$

$$r(t) = A_1 + A_2 t + A_3 \frac{t^2}{2}$$

$$e_{ss} = 0 + 0 + \frac{4}{100} \times 48$$

$$e_{ss} = 1.92$$

8 A system is given by D.E $\frac{d^2y}{dt^2} + \frac{4dy}{dt} + 8y = 8x$ where 'y' is the output and 'x' is input. Determine all time domain specifications for unit step input.

Sol:- Given $\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 8y = 8x$

$$s^2 Y(s) + 4sY(s) + 8Y(s) = 8X(s)$$

$$\frac{Y(s)}{X(s)} = \frac{8}{s^2+4s+8}$$

Given unit step

$$R(s) = \frac{1}{s}$$

by comparing above eqn with standard eqn

$$\frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$$

$\Rightarrow$ $\omega_n^2 = 8$, $2\xi\omega_n = 4$

$\omega_n = \sqrt{8}$, $\xi = \frac{4}{2 \times 2.83}$

$\omega_n = 2.83$, $\xi = 0.707$.

Time domain specifications for unit step

$$t_p = \frac{\pi}{\omega_n\sqrt{1-\xi^2}} = \frac{\pi}{2.83\sqrt{1-(0.7)^2}} = 1.573$$

$$M_p = e^{-\xi\pi/\sqrt{1-\xi^2}}$$

$$= e^{-0.707\times\pi/\sqrt{1-(0.7)^2}}$$

$$= 4.325.$$

t_s :

For 2% tolerance : $\frac{4}{\xi\omega_n} = \frac{4}{0.707\times 2.83} = 1.99 \cong 2s.$

5% tolerance : $\frac{3}{\xi\omega_n} = \frac{3}{0.707\times 2} = 1.493 \cong 1.5s.$

$$t_r = \frac{\pi - \tan^{-1}\left(\frac{\sqrt{1-\xi^2}}{\xi}\right)}{\omega_n\sqrt{1-\xi^2}} = \frac{\pi - \tan^{-1}\left(\frac{\sqrt{1-(0.7)^2}}{0.7}\right)}{2.83\sqrt{1-(0.7)^2}}$$

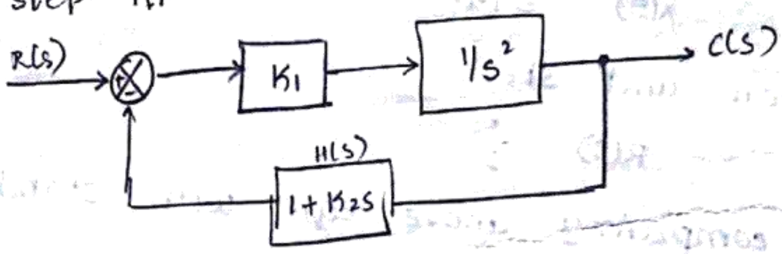
$$= 1.175$$

$$\omega_d = \omega_n\sqrt{1-\xi^2}$$

$$= 2.83\sqrt{1-0.7^2} = 2.0183 \text{ rad/s.}$$

9. For a control system shown in fig. Find the values of K_1 and K_2. so that $M_p = 25\%$ and $t_p = 4s$. assume unit step i/p.

Sol:-



$$G(s) = K_1 \cdot \frac{1}{s^2} = \frac{K_1}{s^2}$$

$$\frac{C(s)}{R(s)} = \frac{G}{1+GH} = \frac{K_1/s^2}{1+\frac{K_1}{s^2}(1+K_2 s)}$$

$$= \frac{K_1}{s^2+K_1(1+K_2 s)} = \frac{K_1}{s^2+K_1+K_1K_2 s}$$

$$= \frac{K_1}{s^2+K_1K_2 s+K_1}$$

compare the eqn with standard eqn $\frac{\omega_n^2}{s^2+2\xi\omega_n s+\omega_n^2}$.

$\Rightarrow \omega_n^2 = K_1$

$\omega_n = \sqrt{K_1}$

$\Rightarrow K_1 = \omega_n^2$

$2\xi\omega_n = K_1K_2$

$2\xi = \frac{K_1K_2}{\sqrt{K_1}}$

$\xi = \frac{K_1K_2}{2\sqrt{K_1}}$

Given

$M_p = 25\%$, $t_p = 4s$

$e^{-(\xi\pi/\sqrt{1-\xi^2})} = M_p$

$e^{-(\xi\pi/\sqrt{1-\xi^2})} = 0.25$

$-(\xi\pi/\sqrt{1-\xi^2}) = \ln(0.25)$

$-(\xi\pi/\sqrt{1-\xi^2}) = -1.39$

$\xi/\sqrt{1-\xi^2} = \frac{1.39}{\pi} = 0.441$

$\xi = \sqrt{1-\xi^2}\,(0.441)$

$\xi^2 = (1-\xi^2)(0.441)^2$

$\xi^2(1+0.0441^2) = (0.441)^2$

$\Rightarrow \xi^2 = 0.169$

$\Rightarrow \xi = 0.403$

$t_p = \frac{\pi}{\omega_n\sqrt{1-\xi^2}} \Rightarrow 4 = \frac{\pi}{\omega_n\sqrt{1-\xi^2}}$

$\Rightarrow 4\omega_n\sqrt{1-\xi^2} = \pi$

$16\omega_n^2(1-\xi^2) = \pi^2$

$\omega_n^2 = \frac{\pi^2}{16(1-\xi^2)}$

$= \frac{\pi^2}{16(1-0.4^2)}$

$\Rightarrow \omega_n = 0.857$

$\omega_n = \sqrt{K_1}$

$\omega_n^2 = K_1$

$K_1 = 0.734$

$\xi = \frac{K_1K_2}{2\sqrt{K_1}}$

$\sqrt{0.4} \times 2 \times \sqrt{0.734} = 0.734\,K_2$

$K_2 = 1.47$

10 A unity feedback system has $G(s) = \frac{40(s+2)}{s(s+1)(s+4)}$.

Determine type of the system, all error coefficients & erro for ramp i/p with magnitude-4.

Sol:- $G(s)H(s) = \frac{40(s+2)}{s(s+4)(s+1)}$ = Type-1 system

$$\frac{G(s)}{R(s)} = 40(s+2)$$

$$K_p = \underset{s\to 0}{Lt}\ G(s)H(s)$$

$$= \underset{s\to 0}{Lt}\ \frac{40(s+2)}{s(s+1)(s+4)} = \infty$$

$$K_v = \underset{s\to 0}{Lt}\ s.\frac{40(s+2)}{s(s+1)(s+4)} = \frac{40\times 2}{4} = 20$$

$$K_a = \underset{s\to 0}{Lt}\ \frac{s^2\, 40(s+2)}{s(s+1)(s+4)} = 0$$

ess for ramp input, $R(s) = 1/s^2$, $r(t) = 4t$.

$R(s) = 4/s^2$.

$$e_{ss} = \underset{s\to 0}{Lt}\ \frac{4s \cdot R(s)}{1+G(s)H(s)}$$

$$= \underset{s\to 0}{Lt}\ \frac{4s/s^2}{1+\frac{40(s+2)}{s(s+1)(s+4)}}$$

$$= \underset{s\to 0}{Lt}\ \frac{4/s}{\frac{s(s+1)(s+4)+40(s+2)}{s(s+1)(s+4)}}$$

$$= \underset{s\to 0}{Lt}\ \frac{(s+1)(s+4)\,4\times 4}{s(s+1)(s+4)+40(s+2)}$$

$$= \frac{4\times 4^2}{40\times 2} = \frac{1}{20} = 0.05\times 4 = 0.2.$$

(or)

$$e_{ss} = \frac{A}{K_v}$$

Given $A = 4$,

$$e_{ss} = \frac{4}{20} = 0.2.$$